

Document: Feature Requirement Document (FRD)

Target Systems: Responsive Web Portal & Mobile Application (Backend Staff/Admin) & Mobile Application (Technicians-Android).

The system consists of Three primary interfaces interacting with a centralized database: **Backend Staff -Web portal +App / Admin -Web portal +App / Customer App / Technician App.**

Project Overview

The objective is to develop a dual-application ecosystem (1.1-Customer App and 1.2-Company Admin App & web + Company app for all Employees (3 Categories: 1.3- Office Backend -Mobile App + Web, 1.4-Field Technicians – Mobile App, 1.5- Sales Executives App for Sales Tracking))

1.1 – Customer App – Login with Registered Mobile with otp

Features required:

- Option to raise complaint/ticket- ticket to be landed to Company website for backend + admin, tracking of ticket & status of ticket.
- New Product purchase options in buy back + Referrals for new sales & service
- Customer app Serves for Subscription & for complete purchase of brand new product. For product purchase e-commerce interface (website link) will be displayed in App.
- User Authentication: OTP-based login (linked to WhatsApp/SMS).
- Live Dashboard: Model Name display, purchase date, Warranty(Residential vs.Commercial).
- Subscription Management: *One-click renewals with instant payments.
- Multi-Brand Marketplace: An e-commerce interface to browse and buy various purifier brands and filter variants. (Separate link with Logo to be embedded in the App – which will take to website where customers can purchase the products).*****
- 1-year warranty tracking and notifications for Complete Product Purchase.
- Payment Gateway: Integrated interface for instant recharges & purchase of complete product.
- Timely notifications of all updates in both WhatsApp & SMS:

Notification types:

- Subscription notification
- App Login otp
- Post Subscription, payment link + payment value + Terms & Conditions
- Product Delivery update with Date
- Post Delivery, Message to select installation date & time (Link) + Quick Start Guide.
- Once Payment done, Customer will receive a Ticket number with Schedule date for installation
- On Installation Date, auto push a Video (FAQ's Video) link notification with on-board message.

Key Notes:

- Once customer subscribes, customer will get a notification in WhatsApp regarding delivery date & installation date with Order id, payment link & Terms and conditions link.

- Missed ivr call from our team to customer, customer should get a notification to contact us back

Marketing & Automation Logic:

- Lead Capture: Any customer who "taps" or clicks a product in the app must trigger an immediate automated WhatsApp message to that user.
- Data Fetching: Customer details (Name, Location, Interest) must be fetched and sent to the Company App/WhatsApp instantly upon interaction.

1.2 – Company Admin App + Web page

- Complete data display of all customer base against Customer id/ customer registered no.
- Option to Create New Logins for backend & Field Technicians (Contact number is login id- which will be logged with otp via SMS)
- Option to assign customer data to Backend with a filter of Pin-code allocation or Model allocation.
- Assign Each backend with a set of Technician ids for tracking technicians
- Complete dash board with options to download an excel report for data download of Calls closures, sales, technician wise & Backend wise.
- Core Functions:
- Master Dashboard: Overview of all active, inactive, and "stopped" devices.
- Granular User Control: Ability to manually search a user and change:

Order Management:

- Instant notification for every new purchase.
- Generation of unique Order IDs for each and every new Subscriber & New Product Purchase.
- Automated delivery of order details to the Company via WhatsApp API.
- Admin must have the power to create "Custom Plans" for high-volume users on the fly.

1.3 – Backend App + Web page

- Create New Unique Customer id for every new purchase & store in cloud
- Create New Service tickets against Unique customer id
- Assign tickets to Technicians with slot & date, follow up tracking system to check whether ticket completed or not.
- Ticket updates with options of remarks + Postponement of customer slot + date & time.
- Download to excel option complete pending dump, with customer contact numbers
- Click to call option for future developments. (IVR) ***
- Ticket Creation & Allocation: Manual or auto-assignment of service and installation requests to the nearest available technician based on pincode/region.
- Ticket Lifecycle Management: Ability to view, edit, re-assign, or cancel tickets with mandatory cancellation reason fields.
- SLA Alerts: Visual cues (color-coding) for tickets approaching their Service Level Agreement (SLA) deadlines.

1.4 – Technician App

- Job Alerts: Push notifications for newly assigned tickets.
- Ticket Acceptance: Option to "Accept" or "Reject" (requires justification) the ticket.
- Status Transitions: Update task status progressively: Assigned → In-Transit → In-Progress → completed / Pending / Cancel
- Spare Part Consumption: A searchable checklist of spare parts to mark items used during service.
- Mandatory Camera Capture: Integration with the mobile device's camera to upload Live photos before and after the repair/installation. Security Requirement: Gallery uploads must be disabled to prevent fraudulent or pre-existing photo submissions.
- Geotagging: Automatic stamping of GPS coordinates and timestamps onto uploaded media.(Latitude & Longitude to be auto enabled for all photos)

1.5 – Sales App (Similar to Technician login access with small Modifications)

- New Sales Lead Generation with Temp id -> once sale is confirmed & purchased new product – Unique id will be created by backend team.
- Sales target dashboard-> Achieved vs Target.
- Job Alerts: Push notifications for newly assigned tickets.
- Ticket Acceptance: Option to "Accept" or "Reject" (requires justification) the ticket.
- Status Transitions: Update task status progressively: Assigned → In-Transit → In-Progress → completed / Pending / Cancel
- Mandatory Camera Capture: Integration with the mobile device's camera to upload Live photos before and after Product delivery. Security Requirement: Gallery uploads must be disabled to prevent fraudulent or pre-existing photo submissions.
- Geotagging: Automatic stamping of GPS coordinates and timestamps onto uploaded media.

System Overview & User Persona Roles

Persona	Primary Interface	Core Responsibility
Backend Staff / Admin	Web Portal + Mobile App	Allocation, tracking, monitoring, reporting, and master data management.
Technician	Mobile App	Executing field tasks (service, installation, warranty, spares) and updating job status in real-time.

****. Technical Stack Recommendations

Backend: Node.js or Python (FastAPI) for high-speed transaction processing.

Database: PostgreSQL (User data) + InfluxDB (Time-series data for water consumption).

Integrations: WhatsApp Business API, Razorpay/Stripe.